EWB COURSES | AWS INTERNSHIP

AWS VPC PEERING

Private Connectivity Between Two Virtual Private Clouds

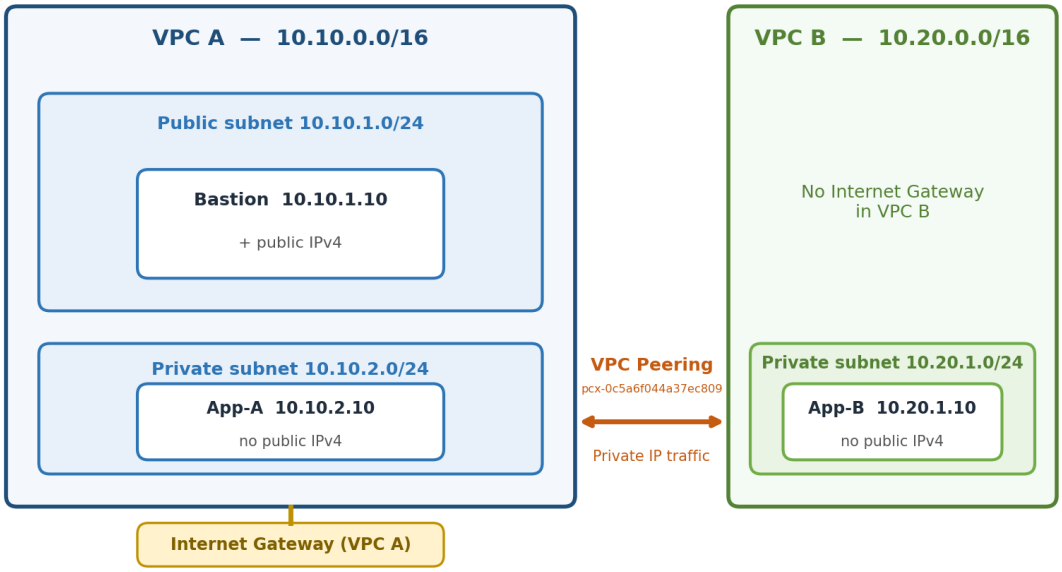
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Batch	AWS Internship Batch 1
Project	EWB Courses — AWS VPC Peering Student Project
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Report date	23 October 2026

1. Executive Summary

The lab builds two non-overlapping Amazon VPC networks and connects them using VPC peering. A public bastion host in VPC A is used for administrative SSH access to two private Amazon Linux application instances. Route tables provide the forward and return paths, while security groups restrict SSH and ICMP traffic to the lab hosts.

The recorded tests demonstrate successful SSH access through the bastion, successful ICMP connectivity in both directions, and a controlled ICMP failure followed by recovery after the rule was restored. Both application instances were shown with no public IPv4 address.

2. Architecture Overview



Architecture: VPC A (bastion + App-A) peered with VPC B (App-B).

Administration path: laptop → bastion public IPv4 → App-A or App-B private IPv4 using SSH ProxyCommand.

Application traffic: App-A ↔ App-B over the peering connection using private IPs.

3. Resource Inventory

The inventory below records resource identifiers that were visible in the AWS console screenshots or confirmed in the conversation. Where a resource ID was not captured, the resource name is provided rather than inventing an ID.

Resource	Name / ID	CIDR / address / notes
VPC A	ewb-vpc-a / vpc-090867e4e55febb1a	10.10.0.0/16
VPC B	ewb-vpc-b / vpc-03430a02270ba249e	10.20.0.0/16
Subnet A public	ewb-a-public / subnet-0a0e428eaa3a344a1	10.10.1.0/24

Resource	Name / ID	CIDR / address / notes
Subnet A private	ewb-a-private / subnet-02d3b403552d20855	10.10.2.0/24
Subnet B private	ewb-b-private / subnet-0a1ef8ea940f73295	10.20.1.0/24
Internet Gateway	ewb-igw-a	Attached to VPC A; full ID not captured
Public route table	ewb-rt-a-public / rtb-093e91f8eb89fb23b	Associated with ewb-a-public
Private route table A	ewb-rt-a-private / rtb-0d023dd71f4bde226	Associated with ewb-a-private
Private route table B	ewb-rt-b-private / rtb-0009e241678eaaffd	Associated with ewb-b-private
VPC peering	ewb-a-b-peer / pcx-0c5a6f044a37ec809	Active
Bastion EC2	ewb-bastion / i-0c7ca71b4d445aa89	10.10.1.10; public IPv4 13.201.97.170 at time of test
App-A EC2	ewb-app-a / i-09ad662eab30919c9	10.10.2.10; no public IPv4
App-B EC2	ewb-app-b / i-03a66c61ab0f0497a	10.20.1.10; no public IPv4
Bastion SG	ewb-sg-bastion / sg-082509ea403f06d62	SSH TCP/22 from laptop public IPv4 /32
App-A SG	ewb-sg-app-a	SSH from 10.10.1.10/32; ICMP Echo Request from 10.20.1.10/32
App-B SG	ewb-sg-app-b / sg-0549d5a0a5bf67cfe	SSH from 10.10.1.10/32; ICMP Echo Request from 10.10.2.10/32
Key pair	ewb-lab	Local private key file: ewb-lab.pem (not included in report)

Instance operating systems: App-A and App-B were confirmed as Amazon Linux 2023. The bastion was observed running Ubuntu 26.04 LTS; its login username is ubuntu. App-A/App-B use ec2-user.

4. Subnets, Route Tables and Routing

The two VPC CIDR ranges do not overlap. Each subnet belongs to its VPC range. The public route table serves the bastion subnet; the two private route tables provide the forward and return paths for application traffic.

Route table	Destination	Target	Purpose / state
ewb-rt-a-public	10.10.0.0/16	local	VPC A local; Active
ewb-rt-a-public	0.0.0.0/0	Internet Gateway	Bastion internet access; Active
ewb-rt-a-public	10.20.0.0/16	pcx-0c5a6f044a37ec809	Bastion to VPC B path; Active
ewb-rt-a-private	10.10.0.0/16	local	VPC A local; Active
ewb-rt-a-private	10.20.0.0/16	pcx-0c5a6f044a37ec809	App-A to App-B; Active
ewb-rt-b-private	10.20.0.0/16	local	VPC B local; Active

Route table	Destination	Target	Purpose / state
ewb-rt-b-private	10.10.0.0/16	pcx-0c5a6f044a37ec809	Return route to VPC A; Active

Subnet associations

Route table	Subnet association
ewb-rt-a-public	ewb-a-public / subnet-0a0e428eaa3a344a1
ewb-rt-a-private	ewb-a-private / subnet-02d3b403552d20855
ewb-rt-b-private	ewb-b-private / subnet-0a1ef8ea940f73295

Routing explanation: a packet from App-A to 10.20.1.10 matches the 10.20.0.0/16 peering route in ewb-rt-a-private. App-B's reply matches the 10.10.0.0/16 peering route in ewb-rt-b-private. Security groups also have to permit the traffic; an Active peering connection alone does not guarantee connectivity.

5. Security Groups and Access

Security group	Inbound rule	Source	Purpose
ewb-sg-bastion	SSH TCP/22	Laptop public IPv4 /32	Restricted administrative access to bastion
ewb-sg-app-a	SSH TCP/22	10.10.1.10/32	SSH from bastion to App-A
ewb-sg-app-a	ICMP Echo Request	10.20.1.10/32	Allow ping from App-B
ewb-sg-app-b	SSH TCP/22	10.10.1.10/32	SSH from bastion to App-B
ewb-sg-app-b	ICMP Echo Request (8/0)	10.10.2.10/32	Allow ping from App-A

The default outbound allow-all rules were retained for this lab. SSH is restricted to the bastion or the user's current public IP /32; SSH and ICMP are not intentionally opened to 0.0.0.0/0. The App-B ICMP rule was temporarily removed for the negative test and then restored.

6. Connectivity Test Results

Test	Command / path	Observed result	Status
Laptop to bastion	SSH to 13.201.97.170 as ubuntu	Ubuntu prompt reached	PASS
Laptop through bastion to App-A	SSH ProxyCommand to 10.10.2.10 as ec2-user	Amazon Linux 2023 prompt reached	PASS
Laptop through bastion to App-B	SSH ProxyCommand to 10.20.1.10 as ec2-user	Amazon Linux 2023 prompt reached	PASS
Forward ping	App-A: ping -c 4 10.20.1.10	4 transmitted, 4 received, 0% loss	PASS
Reverse ping	App-B: ping -c 4 10.10.2.10	4 transmitted, 4 received, 0% loss	PASS
Controlled ICMP failure	App-A ping to App-B after removing App-B ICMP rule	4 transmitted, 0 received, 100% loss	EXPECTED FAILURE

Test	Command / path	Observed result	Status
Recovery	App-A ping to App-B after restoring rule	4 transmitted, 4 received, 0% loss	PASS
Private addressing	EC2 instance details for App-A and App-B	Both show no public IPv4 address	PASS

7. Evidence Screenshots

The screenshots below are selected from the console and terminal evidence shared during the lab. They are included to support the reported configuration and test results.

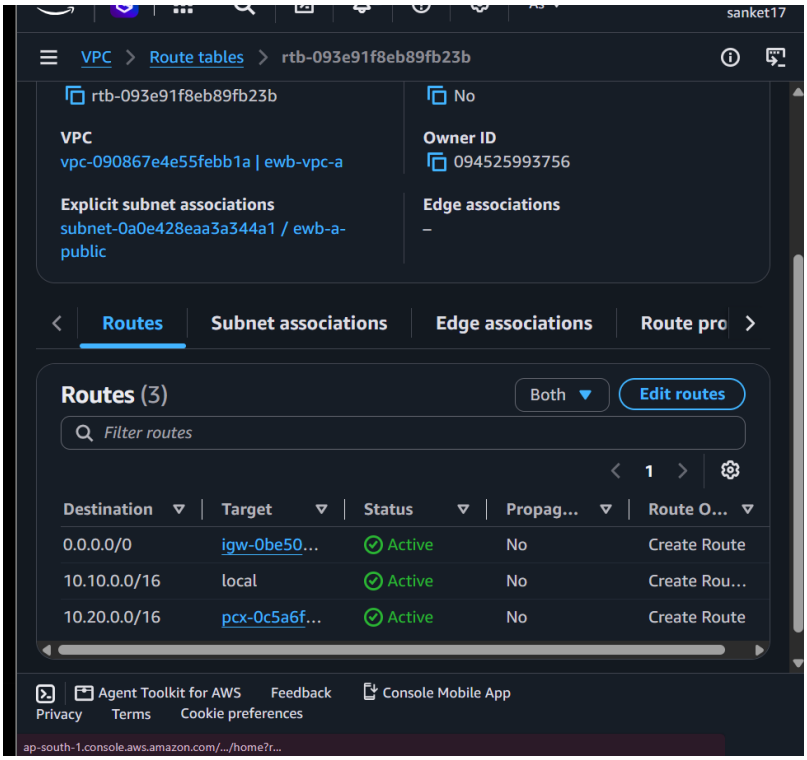


Figure 1. Public route table: local route, Internet Gateway default route, and active peering route.

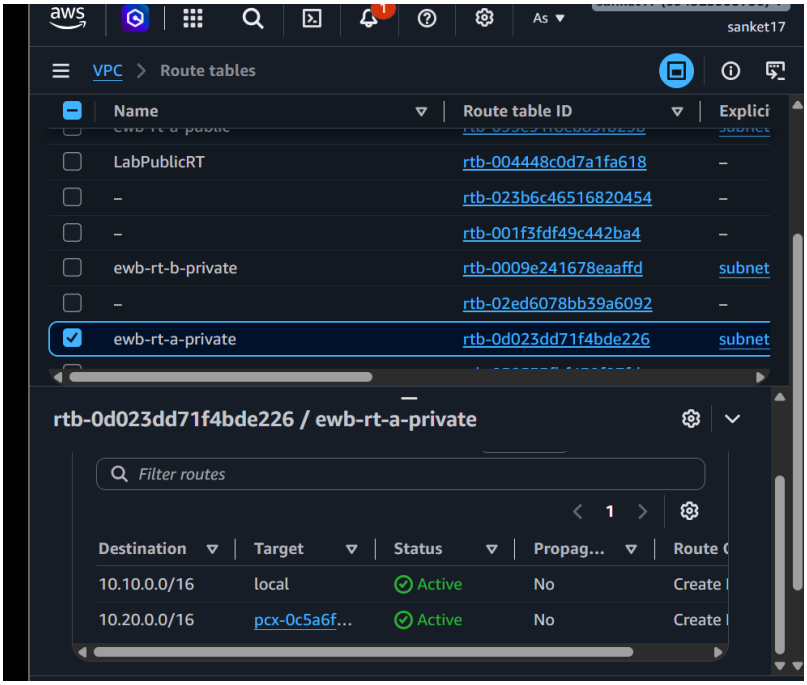


Figure 2. Private route table A showing local route and active route to VPC B through peering.

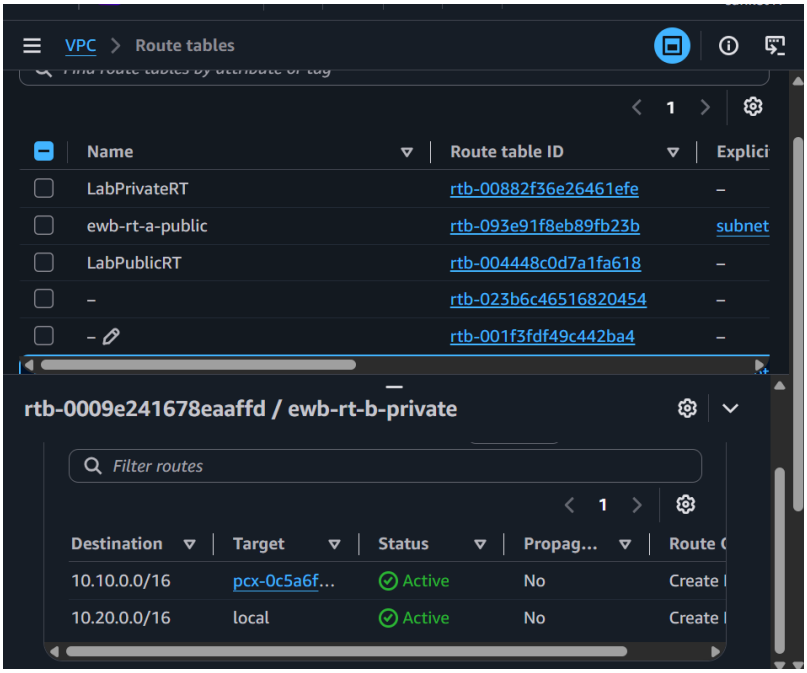


Figure 3. Private route table B showing local route and active return route to VPC A through peering.

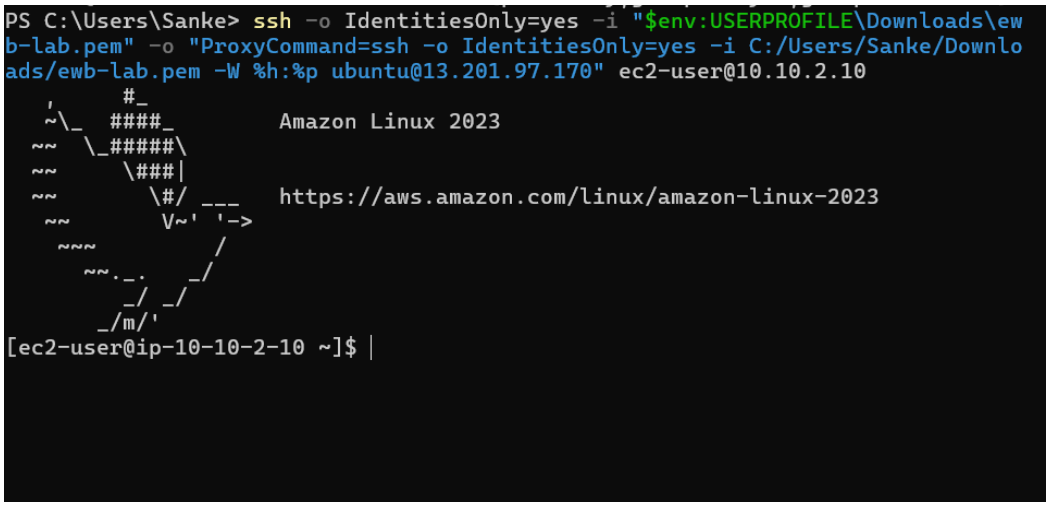


Figure 4. Successful SSH connection to App-A through the bastion; Amazon Linux 2023 prompt.

```
PS C:\Users\Sanke> ssh -o IdentitiesOnly=yes -i "$env:USERPROFILE\Downloads\ewb-lab.pem" -o "
ProxyCommand=ssh -o IdentitiesOnly=yes -i C:/Users/Sanke/Downloads/ewb-lab.pem -W %h:%p ubunt
u@13.201.97.170" ec2-user@10.20.1.10
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:i5OfyIEnVw+1izi05jbqfUvWElZB2cUWOy2pmFMZLeU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$
```

Figure 5. Successful SSH connection to App-B through the bastion; Amazon Linux 2023 prompt.

```
[ec2-user@ip-10-20-1-10 ~]$ hostname
hostname -I
ping -c 4 10.10.2.10
ip-10-20-1-10.ap-south-1.compute.internal
10.20.1.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.191 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.198 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.215 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.212 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3135ms
rtt min/avg/max/mdev = 0.191/0.204/0.215/0.009 ms
[ec2-user@ip-10-20-1-10 ~]$
```

Figure 6. Reverse ping from App-B to App-A: four replies and 0% packet loss.

```
Last login: Thu Oct 22 19:20:02 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.175 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.176 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.216 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.171 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3126ms
rtt min/avg/max/mdev = 0.171/0.184/0.216/0.018 ms
[ec2-user@ip-10-10-2-10 ~]$
```

Figure 7. Forward ping from App-A to App-B: four replies and 0% packet loss.

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3117ms

[ec2-user@ip-10-10-2-10 ~]$
```

Figure 8. Controlled failure: App-B ICMP rule removed; ping from App-A shows 100% packet loss.

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.192 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.246 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.189 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.243 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3089ms
rtt min/avg/max/mdev = 0.189/0.217/0.246/0.027 ms
```

Figure 9. Recovery: ICMP rule restored; ping from App-A to App-B succeeds with 0% packet loss.

8. Additional Evidence and Limitations

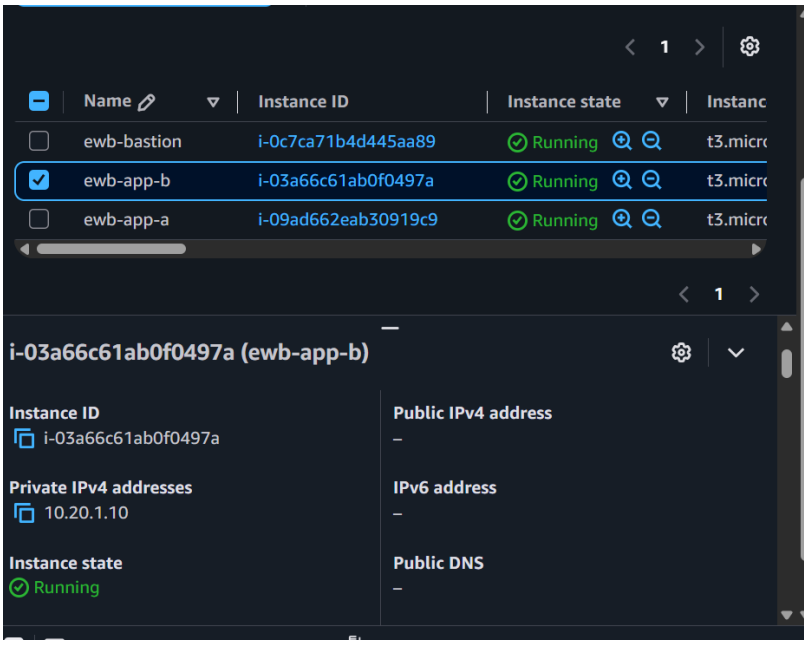


Figure 10. App-B instance details show private IP 10.20.1.10 and no public IPv4 address.

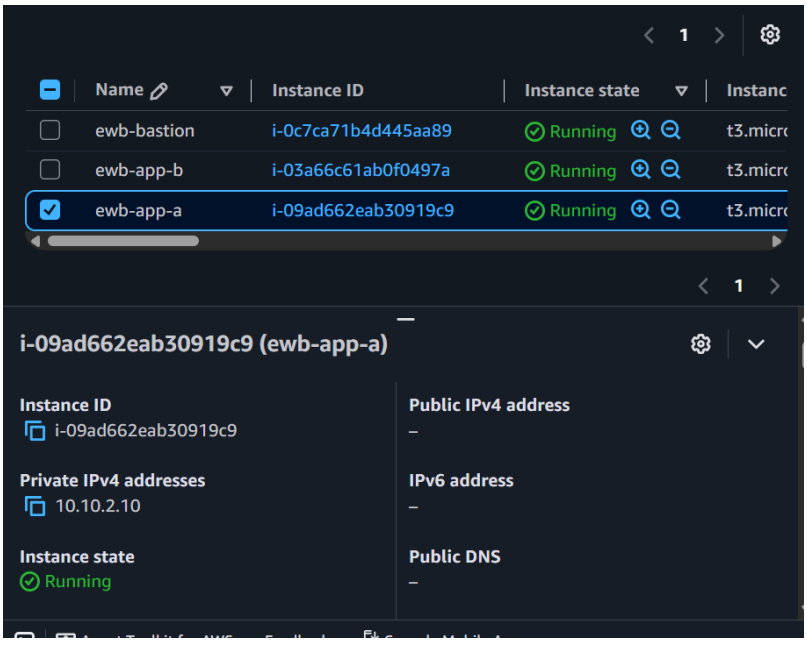


Figure 11. App-A instance details show private IP 10.10.2.10 and no public IPv4 address.

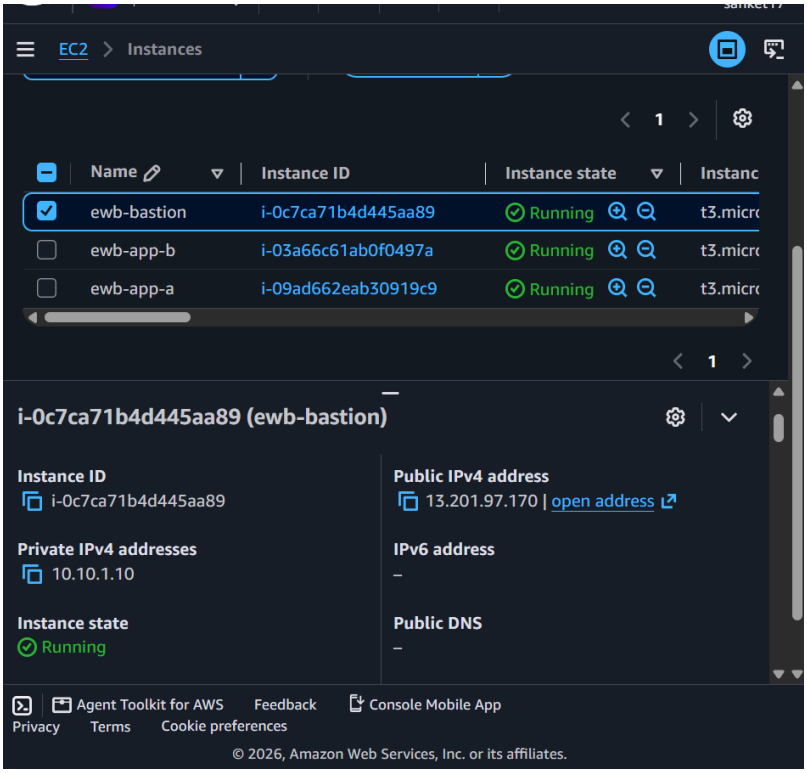


Figure 12. Bastion instance details show private IP 10.10.1.10 and public IPv4 13.201.97.170 at the time of the screenshot.

Evidence still to add before final submission

- A screenshot of the active peering connection showing both VPCs and the peering ID.
- Final inbound rule screenshots for all three security groups, after confirming App-B's SSH and ICMP rules are restored.
- Baseline ping failure before peering routes, if it was not captured earlier.
- Cleanup evidence after lab resources are deleted.

9. Conclusion

The lab demonstrates private communication between two VPCs with non-overlapping CIDRs using VPC peering. The route tables provide forward and return paths, security groups limit inbound SSH and ICMP traffic to the required sources, and the bastion provides administrative access to private application instances. The recorded ping tests prove bidirectional connectivity, and the controlled ICMP test demonstrates that the security group rule affects reachability.

10. Cleanup Checklist — Complete After Saving Evidence

Cleanup has not been evidenced in the screenshots available for this report. Complete this checklist only after the report and evidence have been saved. Terminate the three EC2 lab instances, then detach and delete the Internet Gateway, delete the VPC peering connection, delete custom route tables and security groups where possible, delete the subnets, and finally delete both VPCs. AWS may require resources to be removed in dependency order. Do not delete unrelated resources such as the LabVPC resources visible in the console.

Cleanup action	Status
Save report and evidence screenshots locally	Pending
Terminate ewb-bastion, ewb-app-a, ewb-app-b	Pending
Delete ewb-a-b-peer	Pending
Detach and delete ewb-igw-a	Pending
Delete custom route tables and security groups	Pending
Delete three subnets and both lab VPCs	Pending
Confirm no EWB lab resources remain	Pending

11. What I Learned

- A VPC is an isolated virtual network; subnets divide its address range.
- Route tables decide where packets go; both forward and return routes are necessary.
- VPC peering provides private routing between non-overlapping VPCs, but does not automatically permit traffic.
- Security groups are stateful virtual firewalls; inbound rules can allow or block SSH and ICMP.
- A bastion host provides an administrative entry point to private instances.
- Ping tests and controlled changes provide evidence of both successful connectivity and isolation.